

Wireless LAN, Wi-Fi Standards, and Access Points

A wireless LAN extends the wired campus with radio access points. IEEE 802.11 families, marketed as Wi-Fi, define channels, modulation, and frame formats. Users associate with a service set identifier advertised by one or more access points. Roaming occurs when a client moves and reassociates with a stronger radio.

Coverage planning matters. Overlapping channels on 2.4 GHz cause interference. The 5 GHz and 6 GHz bands offer more non-overlapping channels and higher throughput, especially with MIMO spatial streams. Concrete walls, microwave ovens, and poorly placed antennas create dead zones that no software setting can repair.

Wireless is still a LAN: clients receive DHCP leases, use ARP, and reach the same default gateway as wired neighbors if they share a VLAN. Many organizations place wireless users in a dedicated subnet and filter that subnet more strictly. Guest Wi-Fi should never be bridged directly onto the internal staff VLAN.

Security for wireless LAN includes WPA2 or WPA3, 802.1X enterprise authentication, and rogue access point detection. An open access point in a conference room is a convenient service and a convenient attack surface. Controllers or cloud-managed access points help keep firmware current and SSIDs consistent across buildings.